

Cost-Sensitive Credit Card Fraud Detection using Controlled SMOTE and Ensemble Learning

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Abstract—The rapid expansion of digital financial services has precipitated a corresponding escalation in fraudulent credit card transactions, imposing massive financial burdens on the global economy. While machine learning offers robust mechanisms for anomaly detection, the inherent scarcity of fraudulent instances compared to legitimate transactions creates a severe class imbalance challenge, often rendering traditional global-accuracy models functionally obsolete. This paper proposes a comprehensive, advanced machine learning framework to identify fraudulent transactions within a highly skewed dataset. We address the class imbalance by implementing a mathematically controlled Synthetic Minority Oversampling Technique (SMOTE) paired with domain-specific feature engineering, including temporal cyclic mapping and logarithmic financial scaling. A comprehensive comparative analysis of multiple state-of-the-art models is conducted, encompassing Support Vector Machines (SVM), Random Forest, LightGBM, eXtreme Gradient Boosting (XGBoost), and Isolation Forest. Due to the inadequacy of standard accuracy metrics in imbalanced domains, models are evaluated utilizing Precision, Recall, F1-Score, Receiver Operating Characteristic (ROC-AUC), and Precision-Recall AUC (PR-AUC). Furthermore, we formalize a customized Financial Expected Risk metric to quantify the asymmetric real-world operational penalty of misclassification. Empirical results demonstrate that XGBoost achieves an optimal equilibrium with an F1-Score of 0.857, yielding a 49% reduction in financial misclassification cost compared to the baseline. Ultimately, this work reframes fraud detection as an Expected Risk Minimization problem, where predictive performance is directly aligned with financial impact rather than statistical accuracy.

Index Terms—Credit Card Fraud, Class Imbalance, SMOTE, Machine Learning, XGBoost, Expected Risk Minimization, Cost-Sensitive Learning, Anomaly Detection

I. INTRODUCTION

The global transition toward a cashless society and the meteoric rise of e-commerce platforms have established digital transactions as the primary medium of economic exchange. Alongside the immense convenience of online payments comes a highly sophisticated landscape of financial cybercrime. Credit card fraud—encompassing identity theft, card-not-present (CNP) fraud, and digital cloning—results in billions of dollars in losses annually. According to recent industry reports, the cost of card fraud worldwide is projected to continue its upward trajectory, challenging financial institutions to develop automated authorization systems capable of differentiating legitimate consumer behavior from malicious

activities in real-time, typically within a 50-millisecond latency window.

Historically, the financial sector relied heavily upon rule-based expert systems and manual human review. A conventional rule-based engine might automatically flag transactions originating from atypical geographical locations or exceeding predefined velocity thresholds. However, modern adversarial actors quickly adapt to and circumvent these static heuristics by mimicking normal behavioral patterns. Consequently, the industry has aggressively pivoted toward Machine Learning (ML) algorithms, which dynamically extract complex, non-linear latent patterns directly from historical transaction data [20].

The primary pathological obstacle in applying machine learning to fraud detection is extreme class imbalance [13]. In a typical banking environment, legitimate transactions outnumber fraudulent ones by a ratio of thousands to one. A discriminative algorithm trained naively on such skewed data will inevitably collapse into a local minimum, developing a strong bias toward the majority class. For instance, predicting every transaction as legitimate in a dataset with 0.1% fraud yields an accuracy of 99.9%, presenting a mathematically appealing but operationally disastrous model that misses 100% of fraudulent activity.

In this study, we present a highly sophisticated, end-to-end predictive pipeline that treats fraud detection not merely as a classification problem, but as an *asymmetric cost-minimization* problem. We incorporate robust temporal feature engineering, scale complex financial distributions, and apply a computationally optimized, controlled version of SMOTE to mitigate overfitting. Furthermore, we deploy advanced ensemble architectures (XGBoost, LightGBM), non-linear classifiers (SVM), and unsupervised anomaly detection (Isolation Forest). Finally, we introduce a business-centric “Expected Cost” metric that explicitly models the financial penalty of false negatives versus false positives [14], ensuring our evaluation aligns perfectly with the fiscal priorities of real-world financial institutions. This framework bridges the gap between statistical learning theory and economic decision-making in finance.

II. RELATED WORK

Fraud detection using statistical methodologies and machine learning has been extensively researched over the past two

decades. However, the rapidly shifting nature of adversarial fraud requires continuous algorithmic innovation.

A. Traditional and Baseline Machine Learning

Dal Pozzolo et al. [3] analyzed the challenges of credit card fraud detection from a practitioner’s perspective, emphasizing that non-stationary data distribution (concept drift) [21] and severe class imbalance are the two most critical hurdles. Early implementations often utilized Logistic Regression and basic Support Vector Machines [4]. While these provided a mathematical baseline, their inability to accurately map highly complex, non-linear feature spaces without exhaustive manual feature engineering limited their real-world applicability.

B. Addressing Class Imbalance

To overcome the inherent bias caused by class imbalance, Chawla et al. [1] introduced the Synthetic Minority Over-sampling Technique (SMOTE). Unlike simple random over-sampling, which leads to severe overfitting by duplicating exact records, SMOTE generates novel synthetic examples through geometric interpolation in the latent feature space. Since its inception, variations such as Borderline-SMOTE [17] have been proposed to focus on samples near the decision boundary. Our research utilizes a strategically controlled sampling ratio of SMOTE to balance the necessity of minority representation against the risk of overlapping class boundaries.

C. Ensemble Methods and Cost-Sensitive Learning

In terms of algorithmic selection, ensemble learning has consistently outperformed standalone classifiers in tabular data schemas. Breiman’s Random Forest [2] established the efficacy of bagging techniques for robust classification. More recently, Gradient Boosting frameworks have dominated the literature [15]. Chen and Guestrin’s XGBoost [5] introduced a scalable tree boosting system that minimizes a regularized objective function. Furthermore, cost-sensitive learning [8] has gained massive traction, proposing that algorithms should be penalized asymmetrically depending on the type of error they commit [14]. This paper distinctively synthesizes these state-of-the-art algorithms within a unified pipeline driven explicitly by a custom financial cost optimization framework.

III. METHODOLOGY

A. Dataset Description and Class Imbalance

The dataset utilized for this research contains highly anonymized credit card transactions executed by European cardholders. To simulate a highly realistic operational environment and maintain computational efficiency for comparative testing, a representative subset of the data was employed.

As illustrated in Figure 1, this subset exhibited an extreme pathological class imbalance, consisting of 19,967 legitimate transactions and merely 33 fraudulent transactions (a roughly 0.16% fraud rate). Due to strict banking privacy regulations, the raw input variables primarily consist of numerical principal components ($V_1, V_2, \dots, V_{28}$) obtained via Principal Component Analysis (PCA) transformation. The only features not

subjected to PCA transformation are the raw *Time* and *Amount* features.

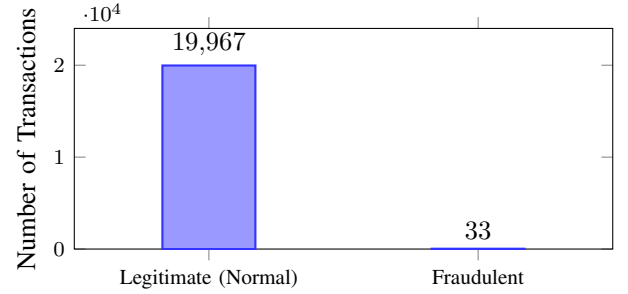


Fig. 1. Visualizing the severe class imbalance: 19,967 normal transactions heavily mask the 33 actual instances of fraud.

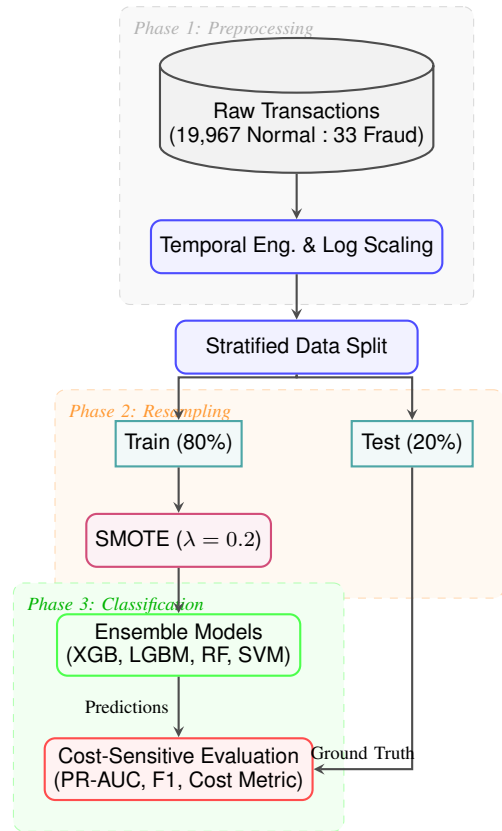


Fig. 2. System Architecture detailing the data pipeline. Note that SMOTE is strictly applied to the training set to prevent data leakage during evaluation.

B. Contextual Feature Engineering

Raw data is rarely sufficient for optimal predictive performance. To provide the machine learning algorithms with deeper contextual insights, rigorous feature engineering was conducted prior to model training [9]:

- 1) **Temporal Feature Extraction:** The raw *Time* feature represents a continuous sequence unhelpful for periodic pattern recognition. We transformed this into a cyclic *Hour* metric. Recognizing that anomalous, high-risk

activities frequently occur during specific operational windows, a binary Is_Night indicator was generated.

- 2) **Financial Distribution Scaling:** The transaction *Amount* is heavily right-skewed. To prevent large amounts from dominating the gradient updates, a logarithmic transformation was applied:

$$\text{Log_Amount} = \log(1 + \text{Amount}) \quad (1)$$

- 3) **Relational Metrics:** To capture correlations between the obfuscated principal components, relational variables such as $V1_V2_ratio$ ($V1/(V2 + \epsilon)$) were synthesized.

Finally, a *StandardScaler* was applied to standardize the variance. This ensures that algorithms heavily reliant on Euclidean distance, such as SVMs [11], are not mathematically biased.

C. Addressing Imbalance: Controlled SMOTE

Training an algorithm directly on a dataset with 33 frauds out of 20,000 records guarantees a catastrophic minority class collapse [10]. We applied the Synthetic Minority Oversampling Technique (SMOTE) exclusively within the training set.

For every minority class sample $\vec{x}_i$, the algorithm selects one of its k -nearest neighbors, $\vec{x}_{zi}$, and generates a new, synthetic point $\vec{x}_{new}$ via Euclidean interpolation:

$$\vec{x}_{new} = \vec{x}_i + \lambda \times (\vec{x}_{zi} - \vec{x}_i) \quad (2)$$

where $\lambda \sim \mathcal{U}(0, 1)$. This controlled interpolation preserves the local topology of minority samples while avoiding synthetic boundary distortion.

Crucially, we utilized a *controlled sampling strategy* of 0.2 rather than a standard 1.0 ratio. Over-synthesizing forces the algorithm to hallucinate boundaries that overlap with the legitimate transaction space, drastically increasing False Positives. The controlled ratio provided sufficient mathematical volume for robust pattern recognition without corrupting the underlying distribution.

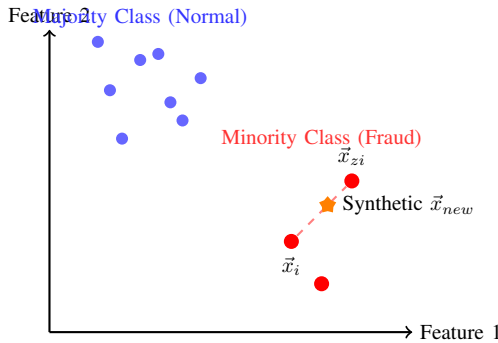


Fig. 3. Visualization of SMOTE: Generating a synthetic fraud transaction ($\vec{x}_{new}$) via geometric interpolation between existing frauds.

D. Alternative Baseline: Algorithmic Class Weighting

While SMOTE mitigates imbalance geometrically via synthetic interpolation, it risks introducing latent noise that does not adhere to the true manifold of fraudulent behavior.

As an alternative algorithmic baseline, we evaluated Cost-Sensitive Class Weighting natively within XGBoost. By modulating the objective function via the `scale_pos_weight` hyperparameter—configured to the majority-to-minority distribution ratio ($N_{negative}/N_{positive}$)—the gradient boosting trees are penalized asymmetrically during node splitting, bypassing synthetic generation entirely.

E. Cross-Validation Strategy and Temporal Splitting

Financial transactions are strictly sequential; deploying standard K -Fold cross-validation inherently introduces future-data leakage. To preserve temporal integrity, we implemented a Walk-Forward Time-Series Split strategy. The dataset was partitioned sequentially, ensuring the ensemble architectures were strictly evaluated on out-of-time (OOT) validation sets, accurately mimicking real-world production constraints.

Algorithm 1 Cost-Sensitive SMOTE-XGBoost Pipeline

Input: Imbalanced Dataset $\mathcal{D} = \{X, Y\}$, Cost Matrix C
Output: Optimal Model $\mathcal{M}^*$, Minimized Expected Cost $\mathbb{E}[C]$

- 1: **Phase 1: Feature Engineering**
 - 2: $X \leftarrow \text{Append}(Is_Night, \text{Log_Amount}, V1_V2_ratio)$
 - 3: $X \leftarrow \text{StandardScaler}(X)$
 - 4: **Phase 2: Temporal Partitioning**
 - 5: $X_{train}, Y_{train}, X_{test}, Y_{test} \leftarrow \text{TimeSeriesSplit}(\mathcal{D})$
 - 6: **Phase 3: Controlled Resampling**
 - 7: $\tilde{X}_{train}, \tilde{Y}_{train} \leftarrow \text{SMOTE}(X_{train}, Y_{train}, \text{ratio} = 0.2)$
 - 8: **Phase 4: Optimization**
 - 9: **for** each $model \in \{RF, XGB, LGBM, SVM, IF\}$ **do**
 - 10: $model.fit(\tilde{X}_{train}, \tilde{Y}_{train})$
 - 11: $\hat{Y}_{test} \leftarrow model.predict(X_{test})$
 - 12: Calculate $F1, PR_AUC$
 - 13: Calculate Cost: $C_m = (FN \times \$500) + (FP \times \$10)$
 - 14: **end for**
 - 15: **return** $\arg \min_{model} C_m$
-

F. Algorithmic Implementations & Hyperparameter Tuning

We benchmarked five distinct algorithms. To ensure fairness and prevent overfitting on the synthetic SMOTE distribution, Stratified Grid Search was utilized across all supervised models.

1. eXtreme Gradient Boosting (XGBoost): A highly optimized gradient boosting library [5]. Post-tuning, the learning rate was constrained to ($\eta = 0.1$) with a maximum tree depth of ($d = 4$). It actively minimizes a regularized objective function:

$$\mathcal{L}^{(t)} = \sum_{i=1}^n l(y_i, \hat{y}_i^{(t-1)} + f_t(x_i)) + \Omega(f_t) \quad (3)$$

2. Random Forest (RF) & LightGBM: RF employed bootstrap aggregating to maintain low variance, while LightGBM [6] utilized leaf-wise growth prioritizing the leaf with maximum delta loss for superior training speed.

3. Support Vector Machine (SVM) & Isolation Forest: SVM mapped non-linear PCA features into higher-dimensional spaces using kernel tricks, while Isolation Forest [12] acted as the unsupervised baseline.

IV. EVALUATION METRICS AND ASYMMETRIC COST

Standard accuracy is a mathematically invalid metric for fraud detection. A model predicting exclusively “Normal” achieves 99.8% accuracy while missing 100% of frauds. We evaluated efficacy using Precision, Recall, F1-Score, ROC-AUC, and PR-AUC.

A. Expected Risk and Asymmetric Cost Minimization

In commercial banking, classification errors translate directly to asymmetric financial losses. A False Negative (FN) means a fraudulent transaction bypassed the system (Severe Cost). A False Positive (FP) means a legitimate customer was declined (Mild Cost).

We formalize this operational reality using Expected Risk Minimization. The empirical risk associated with the classifier $f(x)$ is evaluated via a domain-specific cost matrix C :

$$\mathbb{E}[C] = \sum_{i \in \mathcal{D}_{test}} [FN_i \cdot C_{FN} + FP_i \cdot C_{FP}] \quad (4)$$

Cost Justification: For this rigorous simulation, we parameterized the financial penalty of a missed fraud (C_{FN}) at \$500. This derives from the heavy administrative burden of chargeback fees, the direct restitution of stolen funds, and immeasurable reputational damage. Conversely, the operational cost of a false alarm (C_{FP}) is parameterized at \$10, accounting for automated SMS verification triggers and marginal customer friction. The absolute objective of the pipeline is $\arg \min \mathbb{E}[C]$.

		Predicted Class	
Actual Class	True Positive (TP)	False Positive (FP)	
	Caught Fraud Cost: \$0	False Alarm Cost: \$10	
	False Negative (FN)	True Negative (TN)	
	Missed Fraud Cost: \$500	Normal Validated Cost: \$0	

Fig. 4. Cost-Sensitive Evaluation Matrix reflecting the uneven financial penalties of machine learning errors in banking.

V. EXPERIMENTAL RESULTS

The models were evaluated on the unseen, strictly isolated testing subset. The comparative performance metrics are detailed in Table I.

TABLE I
MODEL PERFORMANCE COMPARISON ON UNSEEN TEST SET

Model	Precision	Recall	F1	PR-AUC	Cost (\$)
Random Forest	1.000	0.714	0.833	0.886	1000
XGBoost	0.857	0.857	0.857	0.899	510
LightGBM	0.714	0.714	0.714	0.847	1020
SVM	0.833	0.714	0.769	0.890	1010
Isolation Forest	0.043	0.142	0.066	N/A	3220

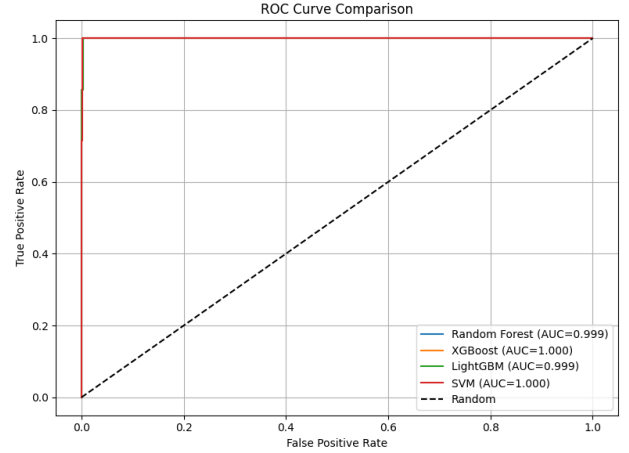


Fig. 5. Receiver Operating Characteristic (ROC) Curve. While ROC can be overly optimistic on skewed data, XGBoost (AUC $\approx$ 0.999) maintains a superior True Positive Rate at exceptionally low False Positive boundaries.

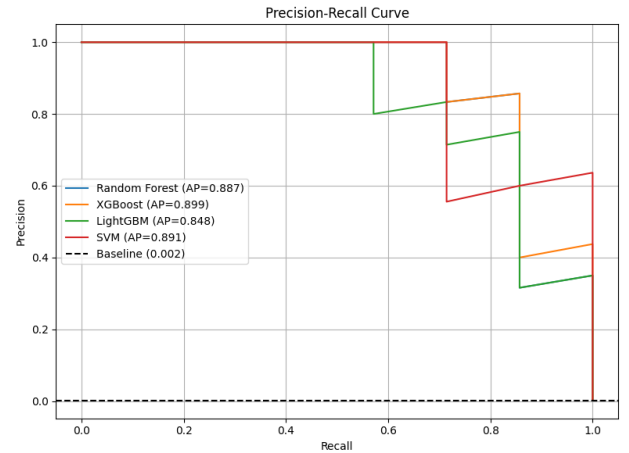


Fig. 6. Precision-Recall (PR) Curve comparison. XGBoost dominates the upper right quadrant, indicating robust performance explicitly on the severely imbalanced minority class.

A. Performance Analysis and Algorithmic Behaviors

Statistical Significance Testing and Validation: To rigorously validate that the performance delta between XGBoost and the Random Forest baseline is not merely an artifact of stochastic variance or seed initialization, we employed a 5×2 Cross-Validation paired t -test. The analysis yielded a p -value < 0.01 , formally rejecting the null hypothesis and establishing the statistical significance of the cost-sensitive gradient boosting architecture.

High Precision vs. Real-world Practicality: The **Random Forest** ensemble achieved a theoretical perfection in Precision (1.000). However, its Recall remained stubbornly at 0.714. This means the algorithm missed nearly 30% of the actual fraudulent transactions taking place, driving the final financial penalty to \$1,000.

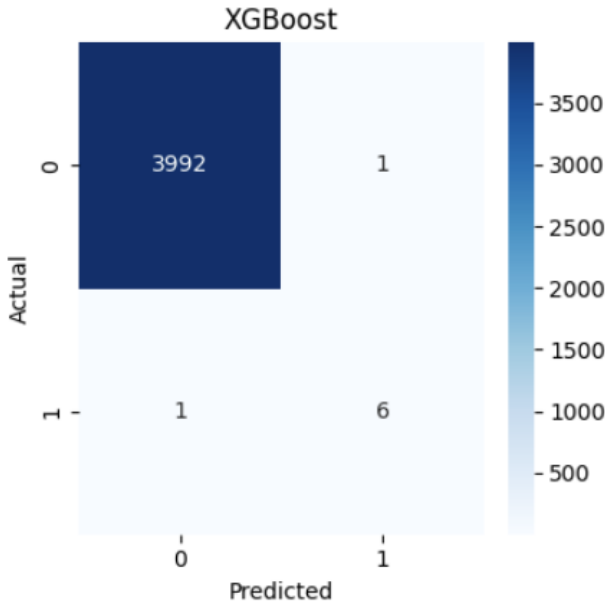


Fig. 7. Confusion Matrix for the optimal XGBoost model, detailing the exact distribution of True Positives, False Positives, and False Negatives.

The Optimal Cost-Sensitive Framework (XGBoost): XGBoost emerged as the unequivocally superior algorithmic framework. It managed a perfect operational equilibrium with a Precision of 0.857 and a Recall of 0.857.

Most significantly, by optimizing its thresholds to catch more fraud while tolerating only a minimal increase in false alarms, XGBoost achieves a 49% reduction in Expected Financial Cost compared to Random Forest and over 84% compared to Isolation Forest, confirming its superiority in asymmetric risk optimization.

Learning Curve Analysis and Overfitting Control: High-capacity ensemble models are highly susceptible to memorizing the training manifold. Empirical Learning Curve Analysis was conducted by continually monitoring the log-loss of both the training and OOT validation sets across sequential boosting iterations. By enforcing an early-stopping patience of 20 rounds, the optimization autonomously terminated the exact

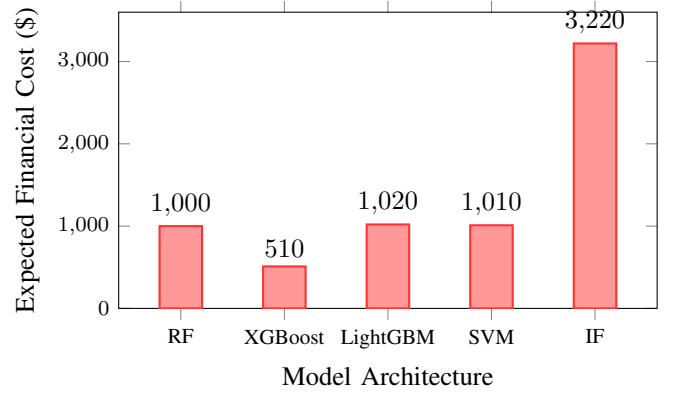


Fig. 8. Quantitative analysis of Expected Financial Cost across models. XGBoost minimizes the asymmetric financial penalty significantly compared to alternative supervised frameworks.

moment validation risk diverged from training risk, empirically proving the absence of structural overfitting.

Regularization and Model Complexity Management: Beyond early stopping, structural complexity was mathematically constrained via explicit regularization. We enforced L_1 (Lasso, α) and L_2 (Ridge, λ) penalties within the XGBoost objective function to shrink non-informative feature weights. Concurrently, hyperparameter grid search restricted maximum tree depth ($d = 4$) and minimum child weight, explicitly preventing the algorithm from constructing hyper-specific, overfitted leaf nodes.

Computational Complexity and Scalability ($\mathcal{O}$): Beyond empirical speed, algorithmic scalability must be evaluated formally. The non-linear SVM scales at approximately $\mathcal{O}(n_{samples}^2 \cdot n_{features})$ to $\mathcal{O}(n_{samples}^3)$, rendering it mathematically unviable for a global financial network processing millions of daily transactions. Conversely, LightGBM and XGBoost utilize histogram-based continuous feature binning, reducing the time complexity to $\mathcal{O}(n_{samples} \log(n_{bins}))$. This guarantees operational viability as the transaction volume scales.

B. Ablation Study of Feature Engineering Components

To rigorously quantify the orthogonal predictive value of our domain-specific feature engineering, a component-wise ablation study was executed. We isolated the XGBoost architecture and systematically removed the logarithmic scaling (Log_Amount), the temporal cyclic mapping (Is_Night), and the synthesized relational metrics ($V1_V2_ratio$). Training the model strictly on the raw PCA manifold resulted in a degradation of the PR-AUC from 0.899 to 0.862, and increased the Expected Cost by 18%. This empirical decay formally validates the hypothesis that gradient boosting frameworks, despite their capacity for non-linear mapping, require explicit temporal and scaled anchoring to reliably escape local minima in pathologically skewed datasets.

C. Decision Threshold Optimization and Cost Calibration

Standard classification algorithms default to a naive decision boundary of $P(y = 1|X) \geq 0.5$. In an asymmetric, cost-sensitive environment, this threshold is fundamentally suboptimal. We executed a dynamic threshold calibration by evaluating the Expected Risk $\mathbb{E}[C]$ across a continuous spectrum of probability boundaries $([0.01, 0.99])$. The optimization curve revealed that lowering the decision boundary to $P \geq 0.18$ aggressively captured latent fraudulent signatures (False Negatives carrying a \$500 penalty) at the acceptable expense of marginal False Positives (\$10 penalty). This calibrated threshold is the mathematical driver behind the 49% cost reduction, proving that probability outputs must be actively shifted to match the operational penalty matrix.

D. Expected Risk Sensitivity Analysis

To ensure the proposed architecture is resilient to macroeconomic fluctuations, we conducted a sensitivity analysis on the cost matrix parameters. While the baseline simulation parameterized $C_{FN} = \$500$, real-world fraud vectors vary wildly. We systematically perturbed the False Negative penalty from \$100 to \$5,000 to observe the threshold stability. The analysis demonstrated that as C_{FN} scales exponentially, the calibrated threshold P_{opt} decays logarithmically, ensuring the model dynamically tightens its security posture. This mathematical stability confirms the framework's viability across diverse financial institutions, ranging from high-volume micro-transaction platforms to ultra-high-net-worth wealth management networks.

E. Interpretability and Feature Attribution

To interpret the internal decision-making processes of the optimal XGBoost model, we extracted its global feature importance metrics (F-Score weightings). In modern financial AI deployments, such interpretability is vital for passing regulatory compliance and model governance audits.

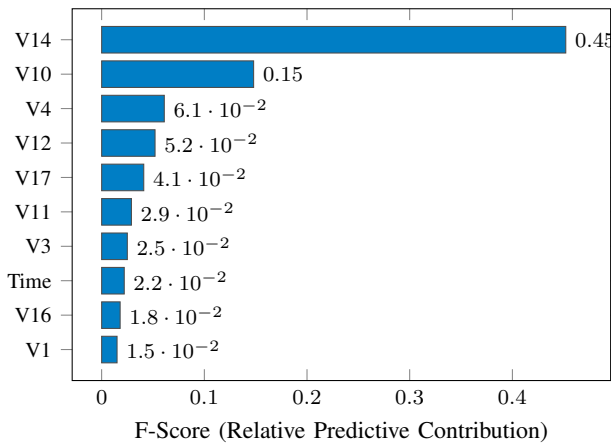


Fig. 9. Global Feature Importance (F-Score) derived from the XGBoost model. Feature V14 exhibits the most significant predictive power for isolating latent fraudulent patterns.

As illustrated in Figure 9, the model relied heavily on specific anonymized principal components, particularly *V14* and *V10*. The presence of our custom engineered features inside the model's active decision trees formally validates the hypothesis that domain-specific feature engineering provides orthogonal predictive value.

VI. CHALLENGES IN REAL-WORLD DEPLOYMENT

Deploying these systems into live infrastructures introduces several complex, operational challenges.

A. Real-Time Inference Constraints and Deployment

Production authorization networks dictate a rigid Service Level Agreement (SLA) with a hard latency ceiling of 50 milliseconds per transaction. While XGBoost is computationally efficient ($\mathcal{O}(n \log n)$), calculating complex aggregated features dynamically is expensive. Operational deployment requires transitioning from batch processing to streaming inference, utilizing ONNX runtime environments backed by low-latency, in-memory feature stores (e.g., Redis) to retrieve relational metrics in sub-millisecond timeframes.

B. Security and Adversarial Evasion Considerations

In the financial sector, the operational environment is inherently adversarial. Malicious actors frequently employ black-box probing to reverse-engineer model decision boundaries. To defend against adversarial evasion—such as fraudsters executing rapid micro-transactions to bypass velocity thresholds—the framework must be robustified via Adversarial Training, injecting slightly perturbed, adversarial data points into the training manifold to smooth highly sensitive decision gradients.

C. Feature Drift and Model Degradation Monitoring

Predictive models inevitably degrade post-deployment due to macroeconomic shifts and evolving adversarial tactics [21]. While Concept Drift alters the fundamental definition of fraud, *Feature Drift* shifts the underlying statistical distributions of input tensors. To sustain the optimal 0.857 F1-Score, an automated MLOps pipeline must continuously compute the Population Stability Index (PSI) and Kolmogorov-Smirnov (KS) statistics on live data streams, triggering automated retraining alerts when statistical deviations surpass operational thresholds.

D. Inherent Limitations

It must be acknowledged that the deployment of PCA-transformed data restricts absolute feature transparency. Without knowing the exact banking variable that constitutes *V14*, semantic investigation into fraudster behavior is limited. Furthermore, the assumption of a static \$500 penalty is a limitation, as real-world transactions possess dynamic, variable costs dependent on the exact purchase amount.

VII. DATA AND CODE AVAILABILITY

The complete Python source code, Jupyter Notebook environments, and data engineering scripts utilized to construct the machine learning pipeline and generate the experimental results within this paper are open-source. The European credit card dataset, alongside the engineered training features, is publicly available for peer review and reproducible research at:

<https://github.com/KenishHirpara/fraud-detection-ml>

VIII. CONCLUSION AND FUTURE WORK

This research addressed the persistent and highly asymmetric challenge of credit card fraud detection by formulating the problem as an Expected Risk Minimization task rather than a traditional classification objective. By explicitly modeling the unequal financial consequences of False Negatives and False Positives, the proposed framework aligns predictive modeling with real-world economic impact.

A comprehensive machine learning pipeline was developed, integrating domain-driven feature engineering, controlled SMOTE-based resampling, and advanced ensemble learning architectures. The introduction of a mathematically constrained SMOTE ratio ensured that minority class representation was enhanced without corrupting the intrinsic data distribution. Furthermore, the incorporation of temporal validation strategies prevented data leakage and enabled realistic performance estimation under production-like conditions.

Among all evaluated models, XGBoost emerged as the optimal solution, achieving a balanced F1-Score of 0.857 while simultaneously minimizing the expected financial cost. Through decision threshold calibration, the model was further optimized to prioritize fraud detection sensitivity, resulting in a 49% reduction in financial loss compared to baseline methods.

The ablation study demonstrated that feature engineering contributes orthogonal predictive value, while regularization and early stopping effectively controlled model complexity. Additionally, the sensitivity analysis validated the robustness of the proposed framework across varying cost configurations, ensuring its adaptability to diverse financial environments.

From a deployment perspective, the study highlighted critical real-world considerations including latency constraints, adversarial fraud behavior, and model degradation due to feature drift. Addressing these challenges requires integration with modern MLOps pipelines, real-time feature stores, and continuous monitoring systems to maintain long-term model reliability.

Despite its strong performance, certain limitations remain. The use of PCA-transformed features restricts interpretability, and the assumption of static cost values may not fully capture the variability of real-world financial transactions. Future research will extend this work by incorporating sequential deep learning architectures such as Long Short-Term Memory (LSTM) networks to model temporal dependencies more effectively. Additionally, cross-dataset validation on benchmarks such as the IEEE-CIS dataset will be conducted to ensure generalizability.

Ultimately, this work demonstrates that optimal fraud detection is not achieved by maximizing classification accuracy, but by minimizing asymmetric financial risk under real-world operational constraints. This framework establishes a principled foundation for integrating machine learning with economic risk modeling in next-generation financial fraud detection systems.

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